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SUBJECT TITLE : PROBABILITY AND STATISTICS

LAB SLOT : L47+L48

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Experiment No. – 3

Binomial, Poisson and Normal Distributions

Mathematics – S.

1. It is known that probability of an item produced by a certain machine will be defective is 0.05. If the produced items are sent to the market in packets of 20, then write down the R code to find the number of packets containing at least, exactly and at most 2 defective items in a consignment of 1000 packets.

Aim

To write an R program to determine the expected number of packets containing exactly 2, at most 2, and at least 2 defective items in a consignment of 1000 packets using the Binomial distribution, where each packet contains 20 items and the probability of a defective item is 0.05.

KEY WORDS

n = Number of trials (items in each packet)
p = Probability of defective item
packets = Total number of packets in the consignment
dbinom() = Function used to compute the probability of exactly k successes in a binomial distribution
pbinom() = Function used to compute the cumulative probability of binomial distribution
exact2 = Variable storing probability of exactly 2 defective items
atmost2 = Variable storing probability of at most 2 defective items
atleast2 = Variable storing probability of at least 2 defective items
exact_packets = Expected number of packets containing exactly 2 defective items
atmost_packets = Expected number of packets containing at most 2 defective items
atleast_packets = Expected number of packets containing at least 2 defective items

R Code

main.r | Share | Run

```r
n=20
p=0.05
packets=1000
exact2=dbinom(2,n,p)
atmost2=pbinom(2,n,p)
atleast2=1-pbinom(1,n,p)
exact_packets=exact2*packets
atmost_packets=atmost2*packets
atleast_packets=atleast2*packets
exact_packets
atmost_packets
atleast_packets
```

Output

```
[1] 188.6768
[1] 924.5163
[1] 264.1605

=== Code Execution Successful ===
```

Conclusion

From the R program using the Binomial distribution:

- Number of packets containing exactly 2 defective items ≈ 188.6768
- Number of packets containing at most 2 defective items ≈ 924.5163
- Number of packets containing at least 2 defective items ≈ 264.1605

Thus, in a consignment of 1000 packets, about 189 packets are expected to contain exactly two defective items, about 925 packets contain at most two defective items, and about 264 packets contain at least two defective items. This demonstrates the application of R programming and binomial probability distribution in quality control analysis.

2. A car hire firm has 2 cars which it hires out day by day. The number of demands for a car on each day follows a Poisson distribution with mean 1.5. Write down the R code to compute the proportion of days on which (i). neither car is used, (ii). at most one car is used and (iii). some demand of car is not fulfilled.

Aim

To write an R program to compute the proportion of days on which

(i) neither car is used
(ii) at most one car is used
(iii) some demand for a car is not fulfilled, assuming the number of daily demands follows a Poisson distribution with mean 1.5.

KEY WORDS

lambda = Mean of the Poisson distribution (average number of daily demands)
dpois() = Function used to compute the probability of exactly x occurrences in a Poisson distribution
ppois() = Function used to compute the cumulative probability of Poisson distribution
neither = Variable storing probability that no car is demanded (X = 0)
atmost1 = Variable storing probability that at most one car is demanded (X ≤ 1)
notfulfilled = Variable storing probability that demand exceeds available cars (X > 2)

R Code

```
lambda=1.5
neither=dpois(0,lambda)
atmost1=ppois(1,lambda)
notfulfilled=1-ppois(2,lambda)
neither
atmost1
notfulfilled
```

```
[1] 0.2231302
[1] 0.5578254
[1] 0.1911532

=== Code Execution Successful ===
```

Conclusion

From the R program using the Poisson distribution (λ = 1.5):

- The proportion of days on which neither car is used is 0.2231302.
- The proportion of days on which at most one car is used is 0.5578254.
- The proportion of days on which some demand for a car is not fulfilled (demand greater than 2 cars) is 0.1911532.

Thus, the Poisson distribution helps determine the probability of different demand situations and analyze whether the available two cars are sufficient to meet daily demand.

3. The local corporation authorities in a certain city install 10,000 electric lamps in the streets of the city with the assumption that the life of lamps is normally distributed. If these lamps have an average life of 1,000 burning hours with a standard deviation of 200 hours, then write down the R code to calculate the number of lamps might be expected to fail in the first 800 burning hours and also the number of lamps might be expected to fail between 800 and 1,200 burning hours.

Aim

To write an R program to calculate the number of electric lamps expected to fail within the first 800 burning hours and between 800 and 1200 burning hours, assuming the lamp life follows a normal distribution with mean 1000 hours and standard deviation 200 hours.

KEY WORDS

mean = Average life of electric lamps
sd = Standard deviation of lamp life
n = Total number of electric lamps installed
pnorm() = Function used to compute cumulative probability in normal distribution
first800 = Probability that lamp life is less than 800 hours
between = Probability that lamp life lies between 800 and 1200 hours
fail_first800 = Expected number of lamps failing before 800 hours
fail_between = Expected number of lamps failing between 800 and 1200 hours

R Code

main.r

```r
mean=1000
sd=200
n=10000
first800=pnorm(800,mean,sd)
between=pnorm(1200,mean,sd)-pnorm(800,mean,sd)
fail_first800=first800*n
fail_between=between*n
fail_first800
fail_between
```

Output

```
[1] 1586.553
[1] 6826.895

=== Code Execution Successful ===
```

Conclusion

From the R program using the Normal distribution (mean = 1000 hours, standard deviation = 200 hours):

- The expected number of lamps that fail within the first 800 burning hours is 1586.553.
- The expected number of lamps that fail between 800 and 1200 burning hours is 6826.895.

Thus, out of 10,000 electric lamps, approximately 1587 lamps are expected to fail before 800 hours and about 6827 lamps are expected to fail between 800 and 1200 hours. This shows how the normal distribution can be used to estimate failure rates and reliability of electrical components.